A Guide To A Career In EMBEDDED SYSTEMS

Embedded systems serve as a bridge between the traditional hardware and software. With artificial intelligence (AI), edge computing, and other advancements, the embedded systems are experiencing meteoric growth with many vast and interdisciplinary sub-fields. So, the term Embedded Systems means different to different people in the industry. This article will help gain some important perspectives about this field and understand the nitty-gritty involved

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Venn diagrams and infographics are a great way to describe the interdisciplinary fields of embedded systems. Venn diagrams depict the nature of embedded systems based on the experiences of different designers. It is interesting to note that most of these diagrams are different from each other in certain aspects, but also have striking similarities.

Some examples are given in the figures that follow.

After looking at the diagrams, you will notice that in many cases embedded sys-

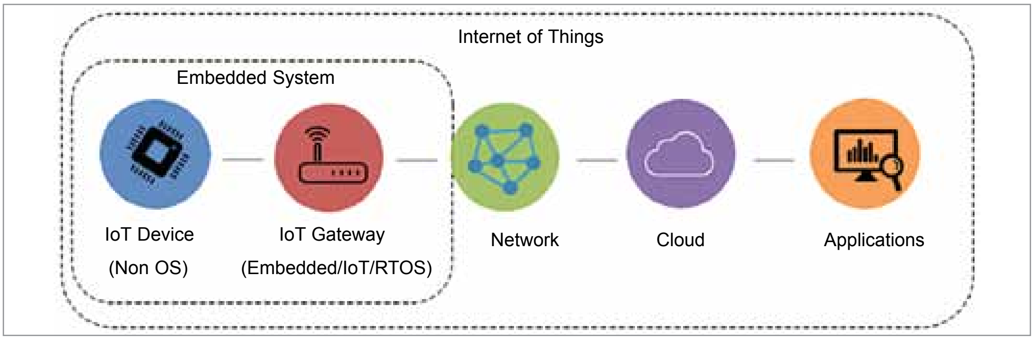

Fig. 1: Embedded system as a subset of IoT (Credit: https://sru.edu.in)

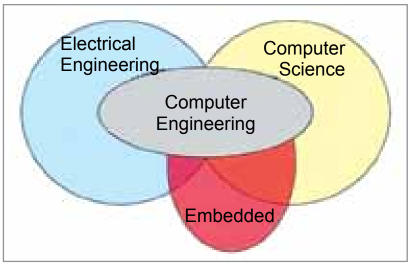

Fig. 2: Embedded evangelist Michael Barr uses this Venn diagram to illustrate the current status of embedded software programming in the computer engineering ecosystem and why there is a lack of good firmware training (Credit: www.eetimes.com)

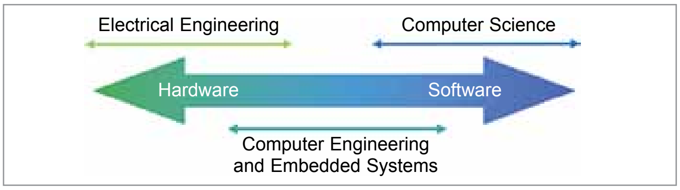

Fig. 3: A description of embedded systems at the TU Delft website (Credit: www.tudelft.nl)

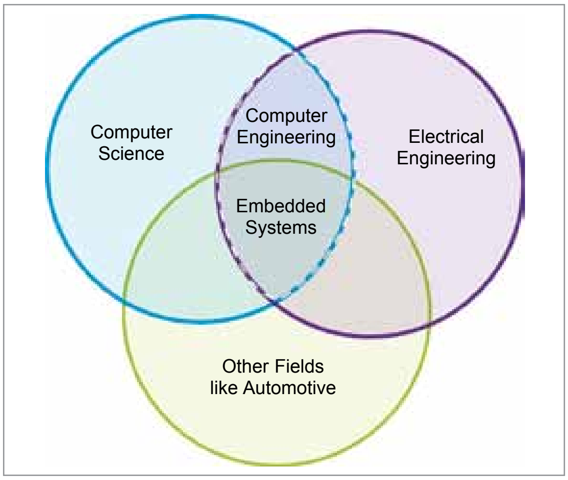

Fig. 4: A Venn diagram that illustrates the three important skills required for embedded systems (Credit: Rosmianto Aji Saputro, https://www.linkedin.com)